Bilkent University Electrical and Electronics Department

EE102-01 Lab 7 Report:

Finite State Machine

Muhammed Mesud Bilirgen – 22301965 – 28th April 2025

Purpose

The aim of this lab was to design and implement a finite state machine (FSM) on a breadboard using D-type flip-flops and basic logic gates.

Design Specification

The FSM designed in this lab has three main states representing emotions: Happy (11), Neutral (01), and Sad (00). An additional unused state (10) was included to ensure that if the system accidentally enters an invalid state (due to noise or glitches), it will safely recover to Neutral. State transitions are as shown in state transition diagram (Figure 1) and in state transition and output table (Table 1).

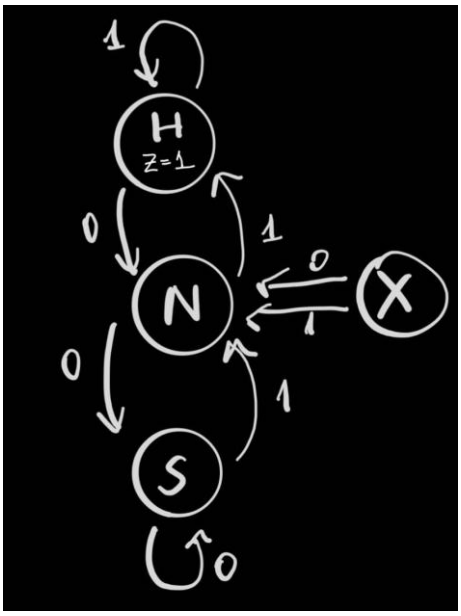

Figure 1: State transition diagram

Present State		Next State		Output
Q_1Q_0	(State)	I: 0	1	Z
00	(Sad)	00	01	0
01	(Neutral)	00	11	0
11	(Happy)	01	11	1
10	(X)	01	01	0

Table 1: State transition and output table

Next state logic equations are derived using Karnaugh maps (Table 2-3).

- $D_1 = Q_0$ AND I
- $D_0 = Q_1$ OR I
- $Z = Q_1$ AND Q_0

The logic equations derived are implemented on a breadboard using D flip-flops, OR gates, AND gates, following the circuit diagram presented in Figure 2. Also, a common clock input is given to D-FFs.

Q_1Q_0	I: 0	1
00	0	0
01	0	1
11	0	1
10	0	0

Table 2: Karnaugh map for D_1

Q_1Q_0	I: 0	1
00	0	1
01	0	1
11	1	1
10	1	1

Table 3: Karnaugh map for D_0

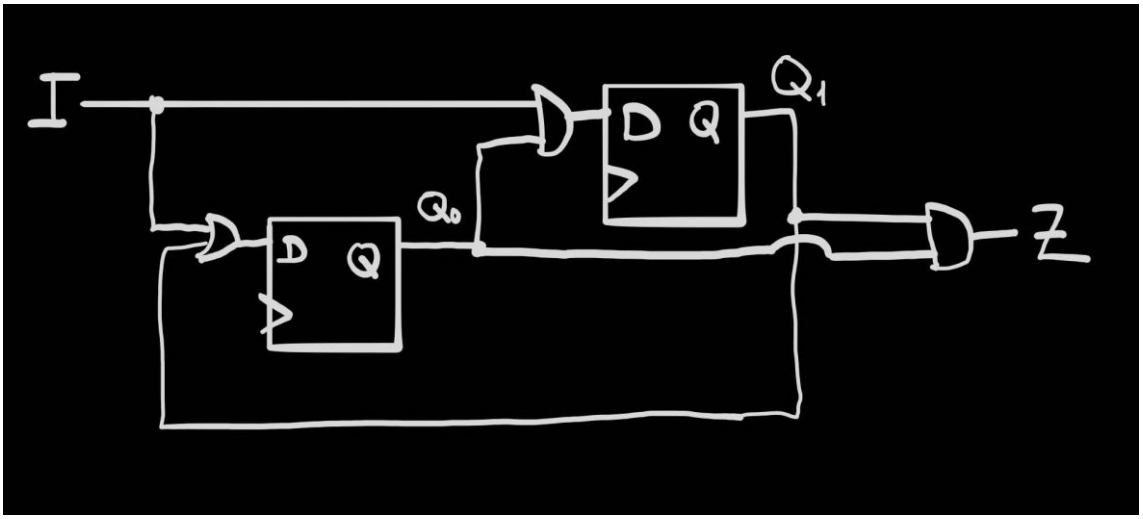

Figure 2: The circuit diagram

Methodology

Datasheets were consulted to understand the functionality and pin configurations of the components. Before starting the experiment, a complete wiring plan was prepared. A 0.1 Hz square wave from the function generator was used as the clock input for the system.

The FSM was realized with two D-type flip-flops (74LS/HC74) to store the state. The input signal was taken from a button circuit. Logic gates 74LS/HC08 (AND) and 74LS/HC32 (OR) were used to generate the next-state logic and the output logic according to the Karnaugh map solutions.

The logic equations D1 = Q0 * I and D0 = Q1 + I were implemented using the available gates. The output Z was generated by Q1 * Q0, and connected to an LED for observation. Additionally, an LED was connected to the clock signal to visualize the clock edges. This setup made it easier to apply the input at the correct time and to test the state transitions of the FSM accurately.

Results

The designed FSM circuit was constructed on the breadboard using two D-type flip-flops (74LS/HC74), OR gates (74LS/HC32), and AND gates (74LS/HC08). A 0.1 Hz square wave from the signal generator was used as the clock source, and a button was used as the input.

An LED was connected to the clock signal to observe clock transitions during operation. Another LED was connected to the output Z to monitor the current state of the system.

The following behavior was observed during testing:

- When starting from the sad state (00), applying an input of '1' transitioned the FSM to Neutral (01).
- From Neutral (01), another '1' input correctly moved the FSM to Happy (11).
- In Happy (11), an input of '0' returned the system to Neutral (01), and another '0' moved it back to Sad (00).
- Output Z was active (logic high) only when the FSM was in the Happy (11) state, consistent with the expected design.
- When the system was forced into the unused state (10) manually, it transitioned correctly to Neutral (01) on the next clock cycle, confirming the recovery mechanism worked as intended.

All observed state transitions and outputs matched the theoretical state transition table and logic equations. Some observations from circuit with yellow Z output LED and red clock LED as shown in Figure 3.1-3.

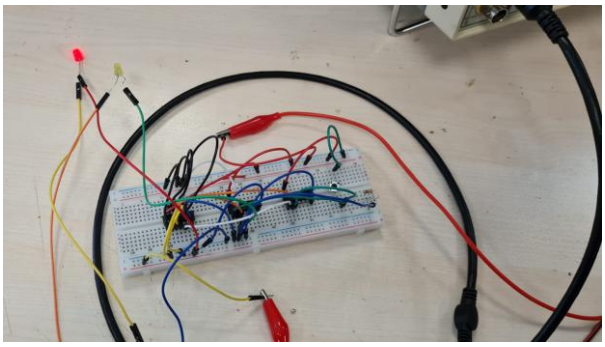

Figure 3.1: The sad state (Z = 0)

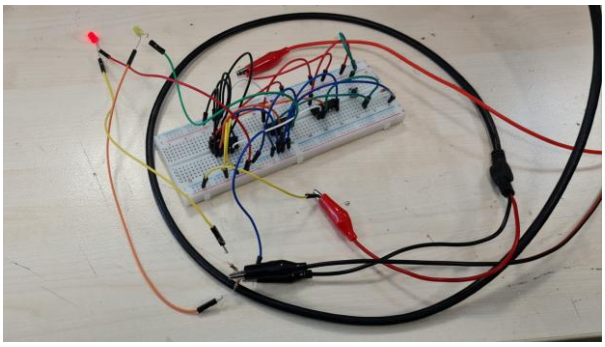

Figure 3.2: The neutral state (Z = 0)

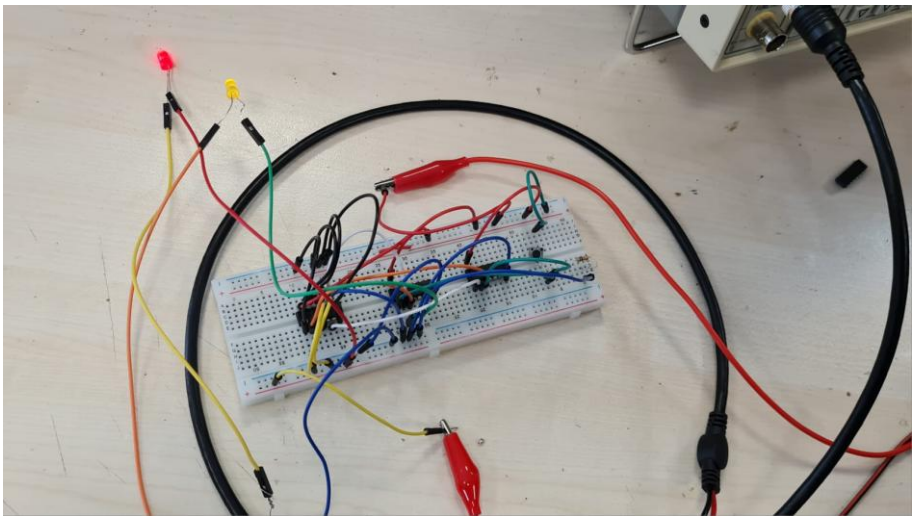

Figure 3.3: The happy state (Z = 1)

Conclusion

In this lab, a finite state machine with three operational states (Sad, Neutral, Happy) and one unused state (X) was successfully designed, implemented, and tested. The FSM transitioned correctly according to the applied inputs, and the output signal Z was generated only when the system was in the happy state, as intended. The logic equations $D_1 = Q_0 * I$ and $D_0 = Q_1 + I$, and the output equation $Z = Q_1 * Q_0$ were correctly implemented using logic gates.

As a result, the experimental results matched the expected behavior, demonstrating a complete and functional FSM system. The experiment successfully reinforced the understanding of finite state machine design, transition logic, and practical circuit implementation.